4-7-26 (WEEK 7)

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MIDTERM SCOPE

Until Week 6 (Not today and after)

SIGMA-ALGEBRAS (Not in the scope of the final)

SETTING

We have infinite seq of tosses of a fair coin. (unif)

$\Rightarrow S = \{H, T\}^{\mathbb{N}}$

QUESTION

What is $\mathbb{P}(\omega)$ $\forall \omega \in S$?

We know fair coin $\Rightarrow$ prob should be "uniform", no one seq is more likely than another

$\mathbb{P}(\omega)$: each toss has prob $\frac{1}{2}$, indep $\Rightarrow$ multiply probabilities

$\therefore \mathbb{P}(\omega) = 0$

We see $S = \bigsqcup_{\omega \in S} \{\omega\}$ is uncountable! (We cannot say $1 = \mathbb{P}(S) = \sum_{\omega \in S} \mathbb{P}(\omega) = 0$) (wrong, no countable additivity)

Defining a "uniform" probability measure

Properties we want:

1. $\mathbb{P}(S) = 1$
2. $\mathbb{P}(\{\omega\}) = 0$ $\forall \omega \in S$
3. If $E_1, E_2, \ldots \subseteq S$ are mutually exclusive, then $\mathbb{P}(\bigsqcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$
4. $\forall i \in \mathbb{N}$, let $T_i : S \to S$ be the map that flips the ith toss
   And for $I \subseteq \mathbb{N}$, let $T_I : S \to S$ be the map that flips the outcomes of all tosses of I
   $\forall E \subseteq S$, $I \subseteq \mathbb{N}$, $\mathbb{P}(T_I(E)) = \mathbb{P}(E)$ (uniform dist)

PROPOSITION

No such $\mathbb{P}$ exists.

Proof

Define an equivalence relation on S:

Say $\omega_1, \omega_2 \in S$ are equivalent if they have finitely many differences, i.e. $\exists n \in \mathbb{N}$, s.t. $(\omega_1)_i = (\omega_2)_i$ $\forall i \geq n$. We say $\omega_1 \sim \omega_2$

Then, we can partition S into equivalence classes $[\omega] = \{\omega' \in S \mid \omega \sim \omega'\}$

Let $E \subseteq S$ be an event that contains one member from all equivalence classes — Axiom of Choice (wtfao idcuh):

Claim: $S = \bigsqcup_{I \subseteq \mathbb{N}, |I| < \infty} T_I(E)$

Proof: $\forall \omega \in S$, we have $[\omega] \cap E = \{\omega'\}$ for some $\omega \sim \omega'$

$\therefore \exists$ finite set of differences $I \subseteq \mathbb{N}$, s.t. $T_I(\{\omega'\}) = \{\omega\}$ $\in T_I(E)$

$\therefore \omega \in \bigcup_{I \subseteq \mathbb{N}, |I| < \infty} T_I(E)$

(Union is disjoint) If $T_I(E) \cap T_J(E) \neq \emptyset$ for some $I \neq J$, $\exists \omega, \omega' \in E$, s.t. $T_I(\{\omega\}) = T_J(\{\omega'\}) = \{\omega''\}$

$\Rightarrow \omega \sim \omega'' \sim \omega' \Rightarrow \omega = \omega'$ since we only have one member of each eq class $\rightarrow\!\leftarrow$

As we have the countable union $S = \bigsqcup_{|I| < \infty, I \subseteq \mathbb{N}} T_I(E)$,

Thus, $1 = \mathbb{P}(S) = \mathbb{P}\left(\bigsqcup_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} T_I(E)\right) = \sum_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} \mathbb{P}(T_I(E)) = \boxed{\sum_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} \mathbb{P}(E)}$

If $\mathbb{P}(E) = 0$, then sum $= 0$, else, sum $= \infty$ $\rightarrow\!\leftarrow$

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DEFINITION

A σ-algebra is a collection of subsets $\Sigma \subseteq 2^S$ satisfying 3 properties:

(1) $S \in \Sigma$

(2) If $E \in \Sigma$, then $E^c \in \Sigma$ (complements)

(3) If $E_1, E_2, \ldots \in \Sigma$, then $\bigcup_{i=1}^{\infty} E_i \in \Sigma$ (countable unions)

Then, we "fix" probability by defining a probability space to be a triple $(S, \Sigma, \mathbb{P})$, where S is the sample space, $\Sigma \subseteq 2^S$ is a σ-alg, and $\mathbb{P}: \Sigma \to \mathbb{R}_{\geq 0}$ assigns a probability to all events (members in Σ)

BOREL σ-ALGEBRA

Σ: generated by intervals $[a, b]$

EXAMPLE

If $|S| < \infty$, then $\Sigma := 2^S$

CUMULATIVE DISTRIBUTION FUNCTION

Given a probability space and a random variable $X: S \to \mathbb{R}$, the cumulative distribution function $F_X: \mathbb{R} \to [0,1]$ is given by $F_X(b) = \mathbb{P}(X \leq b)$

PROPERTIES

① $\lim_{x \to \infty} F_X(x) = 1$, $\lim_{x \to -\infty} F_X(x) = 0$

② $F_X(x)$: increasing for x, i.e. $a \leq b \Rightarrow F_X(a) \leq F_X(b)$

③ Fix n: right-continuous. If $b_n \searrow b$, then $F_X(b_n) \searrow F_X(b)$

Proof

② If $a < b$, then $F_X(b) - F_X(a) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) \overset{\{X \leq a\} \subseteq \{X \leq b\}}{=} \mathbb{P}(a < X \leq b) \geq 0$

$\therefore F_X(a) \leq F_X(b)$ ✓

① Direct result of ②

③ Suppose $b_n \searrow b$. Then, $\bigcap_{n=1}^{\infty} \{X \leq b_n\} = \{X \leq b\}$

$\therefore \lim_{n \to \infty} F_X(b_n) = \lim_{n \to \infty} \mathbb{P}(X \leq b_n) = \mathbb{P}(\underbrace{\lim_{n \to \infty} \{X \leq b_n\}}_{\bigcap_{n=1}^{\infty} \{X \leq b_n\}}) = \mathbb{P}(X \leq b) = F_X(b)$ □

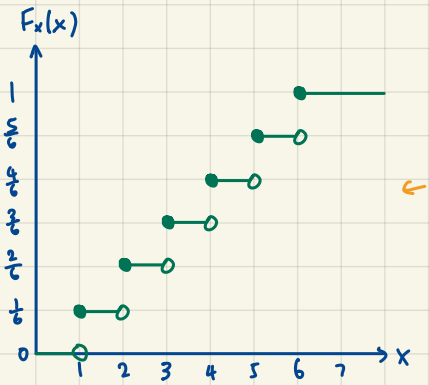
continuity of probability

EXAMPLE

Roll a fair die. X = outcome of roll

Then,

$$F_X(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{6}, & 1 \leq x < 2 \\ \frac{2}{6}, & 2 \leq x < 3 \\ \frac{3}{6}, & 3 \leq x < 4 \\ \frac{4}{6}, & 4 \leq x < 5 \\ \frac{5}{6}, & 5 \leq x < 6 \\ 1, & x \geq 6 \end{cases}$$



← right-conti! Not left-conti

NOTE

F_X does NOT have to be left-continuous but they must be right-continuous

If $b_n \nearrow b$, then $\{X \geq b_n\} \nearrow \{X > b\}$

$\Rightarrow \lim_{n \to \infty} F_X(b_n) = \mathbb{P}(X < b) = F_X(b) - \underbrace{\mathbb{P}(X = b)}_{\text{need not be 0!}}$

DEFINITION

A random variable X is said to be (absolutely) continuous if $\exists$ function $f_X: \mathbb{R} \to \mathbb{R}_{\geq 0}$ s.t. the cummulative distribution function $F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$

Then, f_X is the probability density function of X

In general, $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$. If X is (absolutely) continuous, then it's equal to $\int_{-\infty}^{b} f_X(t)\,dt - \int_{-\infty}^{a} f_X(t)\,dt = \int_a^b f_X(t)\,dt$

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REMARK

What does the density $f_X(t)$ represent?

$\Rightarrow$ A measure of how likely the random variable is to be around x

As $\varepsilon \to 0^+$ for nice enough f_X

$(\because \mathbb{P}(x-\varepsilon \leq X \leq x+\varepsilon) = \int_{x-\varepsilon}^{x+\varepsilon} f_X(t)\,dt \approx 2\varepsilon f_X(x))$

Notice, by the FTC, $f_X(x) = F_X'(x)$

EXAMPLE

Consider

$$f(x) = \begin{cases} \frac{c}{x^3}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

(a) What does c have to be for f to be the pdf of a random variable X?

(b) What is $\mathbb{P}(a \leq X \leq b)$, $1 \leq a \leq b$?

Solution

a) We know $1 = \mathbb{P}(-\infty < X < \infty) = \int_{-\infty}^{\infty} f(t)\,dt = \int_1^{\infty} \frac{c}{t^3}\,dt = \left[-\frac{c}{2t^2}\right]_1^{\infty} = \frac{c}{2} \Rightarrow c = 2$

b) $\mathbb{P}(a \leq X \leq b) = \int_a^b f(t)\,dt = \int_a^b \frac{2}{t^3}\,dt = \left[-\frac{1}{t^2}\right]_a^b = \frac{1}{a^2} - \frac{1}{b^2}$

EXPECTATION (INTUITION)

(Recall) Discrete setting: $\sum x_i \mathbb{P}(X = x_i) = \mathbb{E}(X)$

$\updownarrow$

Continuous setting: $\mathbb{E}(X) := \int_{-\infty}^{\infty} t f_X(t)\,dt$

DEFINITION

If X is a continuous random variable with density function f_X, we define $\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,dt$